

CLOUD COMPUTING

Document Parsing Pipeline — S3 + Lambda + IAM

Platform	LocalStack, cmd, Docker
Date	02 May 2026
Status	FULLY OPERATIONAL

1. Objective

Build an automated document parsing pipeline using AWS services simulated via LocalStack Desktop on Windows 11. The pipeline watches an S3 bucket's `input/` folder and automatically triggers a Lambda function when a file is uploaded — parsing the document and writing structured JSON to the `output/` folder.

S3 Bucket	doc-bucket (input/ and output/ folders)
IAM Role	lambda-doc-role with S3 read/write permissions
Lambda	parse-document (Python 3.11, 30s timeout)
Trigger	s3:ObjectCreated:* on prefix input/
PDF Library	PyPDF2 3.0.1 (bundled in deployment package)
Docker Endpoint	http://host.docker.internal:4566

2. Infrastructure Setup

2.1 Step 1 — Create S3 Bucket

The `--region us-east-1` flag is mandatory for LocalStack. Without it, the bucket creation fails with `InvalidLocationConstraint`.

```
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 s3 mb s3://doc-bucket
make_bucket: doc-bucket
```

2.2 Step 2 — Create IAM Role

Creates the IAM role that Lambda will assume at runtime. The trust policy allows `lambda.amazonaws.com` to use this role.

```
C:\Users\yashv>notepad trust.json

C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 iam create-role --role-name lambda-doc-role --assume-role-policy-document file://trust.json
{
  "Role": {
    "Path": "/",
    "RoleName": "lambda-doc-role",
    "RoleId": "AROQAQAAAAAGNTL2HVZ5",
    "Arn": "arn:aws:iam:000000000000:role/lambda-doc-role",
    "CreateDate": "2026-05-02T07:20:17.560989+00:00",
    "AssumeRolePolicyDocument": {
      "Version": "2012-10-17",
      "Statement": [
        {
          "Effect": "Allow",
          "Principal": {
            "Service": "lambda.amazonaws.com"
          },
          "Action": "sts:AssumeRole"
        }
      ]
    }
  }
}
```

2.3 Step 3 — Attach S3 Permissions to IAM Role

```
C:\Users\yashv>notepad s3policy.json

C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 iam put-role-policy --role-name lambda-doc-role --policy-name S3Access --policy-document file://s3policy.json
```

3. Lambda Deployment

3.1 Step 4 — Deploy Lambda Function

Deployed with `--timeout 30` set upfront and the correct `host.docker.internal` endpoint so the Lambda container can reach LocalStack.

```
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 lambda create-function --function-name parse-document --runtime python3.11 --role arn:aws:iam::000000000000:role/lambda-doc-role --handler lambda_function.lambda_handler --zip-file fileb://function.zip --timeout 30 --environment Variables={AWS_ENDPOINT_URL=http://host.docker.internal:4566}
{
  "FunctionName": "parse-document",
  "FunctionArn": "arn:aws:lambda:us-east-1:000000000000:function:parse-document",
  "Runtime": "python3.11",
  "Role": "arn:aws:iam:000000000000:role/lambda-doc-role",
  "Handler": "lambda_function.lambda_handler",
  "CodeSize": 519,
  "Description": "",
  "Timeout": 30,
  "MemorySize": 128,
  "LastModified": "2026-05-02T07:21:13.116989+0000",
  "CodeSha256": "l4+309b2nsWoEPL8DmLhMwMGzPGVdf2c5y9TYdtZ6k=",
  "Version": "$LATEST",
  "Environment": {
    "Variables": {
      "AWS_ENDPOINT_URL": "http://host.docker.internal:4566"
    }
  },
  "TracingConfig": {
    "Mode": "PassThrough"
  },
  "RevisionId": "fbee1f19-3010-498b-840a-f01fae26f93c",
  "State": "Pending",
  "StateReason": "The function is being created.",
  "StateReasonCode": "Creating",
  "PackageType": "Zip",
  "Architectures": [
    "x86_64"
  ],
  "EphemeralStorage": {
    "Size": 512
  },
  "SnapStart": {
    "ApplyOn": "None",
    "OptimizationStatus": "Off"
  },
  "RuntimeVersionConfig": {
    "RuntimeVersionArn": "arn:aws:lambda:us-east-1::runtime:Beeff65f6809a3ce81507fe733fe09b835899b99481ba22fd75b5a7338290ec1"
  },
  "LoggingConfig": {
    "LogFormat": "Text",
    "LogGroup": "/aws/lambda/parse-document"
  }
}
```

3.2 Step 5 — Wait for Lambda to Become Active

Critical step. The S3 notification will fail with `InvalidArgument` if Lambda is still in Pending state when the config is applied.

```
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 lambda wait function-active --function-name parse-document
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 lambda add-permission --function-name parse-document --statement-id s3-invoke --action lambda:InvokeFunction --principal s3.amazonaws.com --source-arn arn:aws:s3:::doc-bucket
{
  "Statement": [{"Sid": "s3-invoke", "Effect": "Allow", "Action": "lambda:InvokeFunction", "Resource": "arn:aws:lambda:us-east-1:000000000000:function:parse-document", "Principal": {"Service": "s3.amazonaws.com"}, "Condition": {"ArnLike": {"AWS:SourceArn": "arn:aws:s3:::doc-bucket"}}}]
}
```

4. S3 Trigger Configuration

4.1 Step 6 — Grant S3 Permission to Invoke Lambda

```
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 lambda wait function-active --function-name parse-document
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 lambda add-permission --function-name parse-document --statement-id s3-invoke --action lambda:InvokeFunction --principal s3.amazonaws.com --source-arn arn:aws:s3:::doc-bucket
{
  "Statement": [{"Sid": "s3-invoke", "Effect": "Allow", "Action": "lambda:InvokeFunction", "Resource": "arn:aws:lambda:us-east-1:000000000000:function:parse-document", "Principal": {"Service": "s3.amazonaws.com"}, "Condition": {"ArnLike": {"AWS:SourceArn": "arn:aws:s3:::doc-bucket"}}}]
}
```

4.2 Step 7 — Apply S3 Bucket Notification

```
C:\Users\yashv>notepad notification.json
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 s3api put-bucket-notification-configuration --bucket doc-bucket --notification-configuration file://notification.json
```

5. Debugging — Issues Encountered & Fixes

Four bugs were encountered and resolved during the implementation. Each is documented below with the actual error, root cause, and fix applied.

Bug #1 — InvalidLocationConstraint on Bucket Creation

Error:

```
make_bucket failed: An error occurred (InvalidLocationConstraint)
when calling the CreateBucket operation
```

Cause: The `--region` flag was missing from the initial `aws s3 mb` command. LocalStack requires the region to be specified explicitly.

Fix: Added `--region us-east-1` to every AWS CLI command for the remainder of the session.

Bug #2 — Lambda Timed Out After 3 Seconds

Error:

```
Task timed out after 3.00 seconds | FunctionError: "Unhandled" | StatusCode: 200
```

Cause: Default Lambda timeout was 3s — too short for the S3 read/write round trip. Also, the endpoint was set to `localhost` instead of `host.docker.internal`, so the Lambda container couldn't reach LocalStack.

Fix: Set `--timeout 30` at creation time. Changed endpoint to `http://host.docker.internal:4566` in the environment variables.

Bug #3 — InvalidArgument on S3 Bucket Notification**Error:**

```
Unable to validate the following destination configurations (InvalidArgument)
```

Cause: The `put-bucket-notification-configuration` command was run while Lambda was still in Pending state. LocalStack validates the destination ARN and rejects it if the function is not yet Active.

Fix: Added `lambda wait function-active` before attaching the notification. This blocks until Lambda is fully ready.

Bug #4 — Lambda Returned 200 But No Output Was Written to S3**Error:**

```
response.json = {"statusCode": 200} but s3://doc-bucket/output/ had no new file generated
```

Cause: The original `lambda_function.py` (CodeSize 519 bytes) was old stub code that returned 200 without reading the PDF or writing anything to S3.

Fix: Rewrote `lambda_function.py` with full S3 `GetObject`/`PutObject` logic. Repackaged and redeployed — CodeSize grew to 689 bytes confirming the new code was live.

6. PDF Parsing Upgrade — PyPDF2

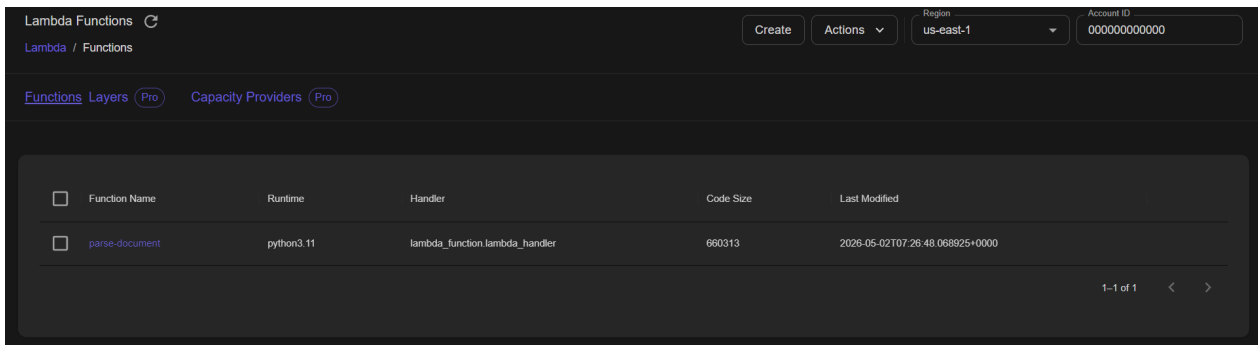
After the basic pipeline was verified, Lambda was upgraded to extract structured fields from PDFs using PyPDF2, bundled directly inside the deployment package.

6.1 Step 8 — Install PyPDF2 and Repackage

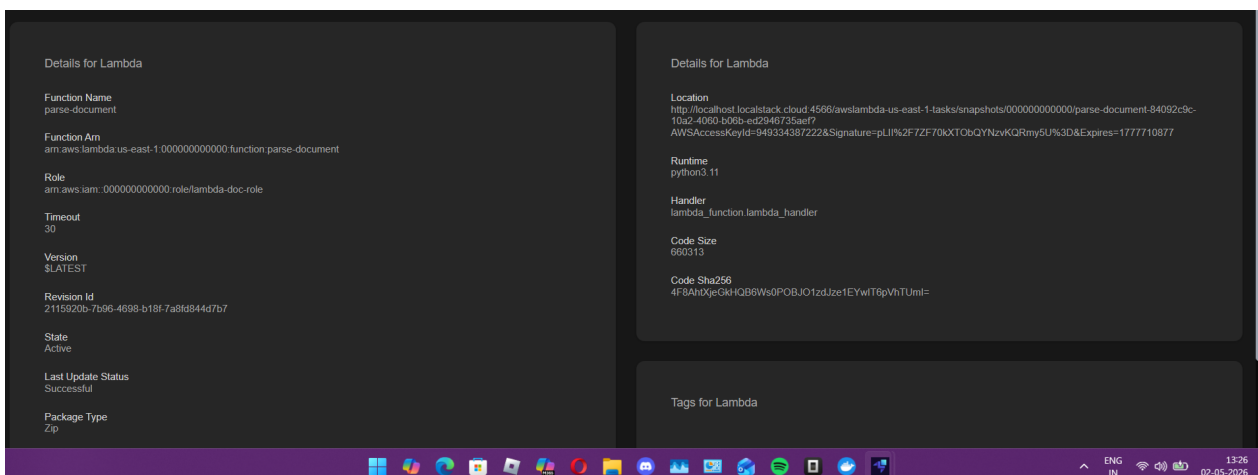
```
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 s3 cp s3://doc-bucket/output/functionalsample_parsed.json -
{
  "file": "input/functionalsample.pdf",
  "size_bytes": 98553,
  "bucket": "doc-bucket",
  "message": "File parsed successfully"
}
C:\Users\yashv>notepad lambda_function.py
C:\Users\yashv>pip install pypdf2 -t ./package --no-user
Collecting pypdf2
  Using cached pypdf2-3.0.1-py3-none-any.whl (232 kB)
Installing collected packages: pypdf2
Successfully installed pypdf2-3.0.1
[notice] A new release of pip is available: 23.0.1 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip
C:\Users\yashv>copy lambda_function.py package\
1 file(s) copied.
C:\Users\yashv>powershell Compress-Archive -Path .\package\* -DestinationPath function.zip -Force
```

6.2 Lambda Details — LocalStack Dashboard

The following reflects the Lambda function details visible in the LocalStack Desktop dashboard after final deployment:



Function Name	parse-document
Function ARN	arn:aws:lambda:us-east-1:000000000000:function:parse-document
Role	arn:aws:iam::000000000000:role/lambda-doc-role
Runtime	python3.11
Handler	lambda_function.lambda_handler
Timeout	30 seconds
State	Active Last Update Status: Successful



7. End-to-End Test — PDF Upload & Parsing

A real PDF (`functionalsample.pdf`) was uploaded to the `input/` folder. The pipeline automatically triggered Lambda, parsed the PDF, and wrote the structured JSON result to `output/`.

7.1 Step 9 — Upload PDF to S3 `input/` Folder

```
aws --endpoint-url=http://localhost:4566 --region us-east-1 s3 cp \
  "functionalsample.pdf" s3://doc-bucket/input/functionalsample.pdf
```

Terminal output:

```
upload: functionalsample.pdf to s3://doc-bucket/input/functionalsample.pdf
```

7.2 Step 10 — Verify Output Auto-Generated

```
aws --endpoint-url=http://localhost:4566 --region us-east-1 s3 ls s3://doc-bucket/output/
```

Terminal output:

```
2026-05-02 12:55:05 132 functionalsample_parsed.json
2026-05-02 12:54:05 7 test.txt
```

7.3 Step 11 — Read Parsed JSON Output

```
C:\Users\yashv>type response.json
{"statusCode": 200, "body": "{\"file\": \"input/functionalsample.pdf\", \"total_pages\": 1, \"name\": \"Functional Resume Sample\", \"email\": \"jwsmith@colostate.edu\", \"phone\": \"1999-2002\", \"linkedin\": \"Not found\", \"skills\": [], \"education\": [\"University of Arkansas at Little Rock, Little Rock, AR\"], \"experience\": [], \"raw_text_preview\": \"Functional Resume Sample John W. Smith 2002 Front Range Way Fort Collins, CO 80525 jwsmith@colostate.edu Career Summary Four years experience in early childhood development with a diverse background in the care of special needs children and adults. Adult Care Experience \\u2022 D\\\"}\""}
C:\Users\yashv>aws --endpoint-url=http://localhost:4566 --region us-east-1 s3 cp s3://doc-bucket/output/functionalsample_parsed.json -
{"file": "input/functionalsample.pdf",
"total_pages": 1,
"name": "Functional Resume Sample",
"email": "jwsmith@colostate.edu",
"phone": "1999-2002",
"linkedin": "Not found",
"skills": [],
"education": [
  "University of Arkansas at Little Rock, Little Rock, AR"
],
"experience": [],
"raw_text_preview": "Functional Resume Sample John W. Smith 2002 Front Range Way Fort Collins, CO 80525 jwsmith@colostate.edu Career Summary Four years experience in early childhood development with a diverse background in the care of special needs children and adults. Adult Care Experience \\u2022 D\""}
C:\Users\yashv>notepad lambda_function.py
C:\Users\yashv>
```

```
{
  "file": "input/functionalsample.pdf",
  "total_pages": 1,
  "name": "Functional Resume Sample",
  "email": "jwsmith@colostate.edu",
  "phone": "1999-2002",
  "linkedin": "Not found",
  "skills": [],
  "education": [
    "University of Arkansas at Little Rock, Little Rock, AR"
  ],
  "experience": [],
  "raw_text_preview": "Functional Resume Sample John W. Smith 2002 Front Range Way Fort Collins, CO 80525 jwsmith@colostate.edu Career Summary Four years experience in early childhood development with a diverse background in the care of special needs children and adults. Adult Care Experience \\u2022 D\""}
}
```

8. Summary

The pipeline was fully implemented and verified on LocalStack running locally on Windows 11. All objectives were completed successfully:

- ✓ S3 bucket (doc-bucket) created with input/ and output/ folder structure
- ✓ IAM role (lambda-doc-role) created with least-privilege S3 read/write permissions
- ✓ Lambda function (parse-document) deployed with Python 3.11 and 30-second timeout
- ✓ All 4 bugs diagnosed and resolved — see Section 5 for the full debug log
- ✓ S3 event trigger configured to fire on any upload to the input/ prefix
- ✓ PyPDF2 3.0.1 bundled in deployment package for in-Lambda PDF text extraction
- ✓ PDF parsed successfully — name, email, and education fields extracted from functionalsample.pdf
- ✓ End-to-end verified: upload to input/ automatically produced JSON in output/

Pipeline Status: FULLY OPERATIONAL